Moral Tensors and DecisionProofs: Compiling Language into an Auditable, Grounded Safety Layer

Andrew H. Bond
San José State University & independent researcher ORCID 0009-0003-2599-6158
PyTorch Conference North America 2026, Poster Session (Responsible AI)

September 2026 (rev. 3: regime-transition worked example; first activation-lens calibration)

Abstract

Most AI safety is a scalar at the output: one good/bad check before an action ships. We keep the structure instead (who is affected, what is owed, who consented, who bears imposed risk) and contract to a decision only at the end, auditably. Two open-source pieces realize this. `erisml-compiler` turns natural-language moral material into a typed MoralGraph and a rank-1…6 moral tensor, read through four ethical lenses at once; when the lenses disagree it refuses to silently aggregate and defers to a human. At runtime, ErisML's three-layer gateway drops the same evaluation into an agent's plan→act loop (Fig. 1), sealing each decision in a SHA-256 hash-chained DecisionProof and failing safe, never open. The geometry grounds it: correlative swap and deontic negation are commuting involutions on Hohfeld's normative positions, generating the *measured* Klein four-group V_4; the full dihedral D_4 remains a posited, testable extension, a correction of our own earlier claim, documented here. A Bond Index tests whether a judgment survives the agent↔patient swap. The PyTorch hook is literal: forward hooks on transformer layers compare what a model *says* with what it internally *exhibits*. Running today on a Qwen/Gemma agent stack and two governed AI NPCs.

1 Structure before the scalar

A single good/bad score discards exactly the information an audit needs: which stakeholders are affected, what is owed to each, who consented, and who bears imposed risk. ErisML keeps that structure as a typed object through the whole evaluation and performs *decision contraction* only at the end, logging the weights used, who lost, and what moral residue remains. The claim is not that the system knows what is good; it is that every collapse from structure to verdict is recorded and replayable.

2 The compiler

`erisml-compiler` (alpha v0.9.0, PyPI, MIT, 477 tests, Zenodo DOI 10.5281/zenodo.20659432) runs a 12-pass pipeline: `text` → `segment` → `extract` → `canonicalize` → `MoralGraph` → `tensorize` → `DEME` → `DecisionProof`. The MoralGraph is typed (stakeholder, act, maxim, commitment, fact, norm nodes) and carries a canonical SHA-256 hash. Extraction has three tiers: deterministic rules, an LLM tier (vLLM with a critic pass), and a LaBSE classifier-head probe.

Framework pluralism. One graph is read through four projections: consequentialist (harm tensor; Gini, worst-off, exact Shapley attribution), deontic (Kantian universalizability gates formulated in Z3 SMT: an UNSAT core witnesses a contradiction in conception), virtue, and care. The projections are not averaged. When they disagree, `cross_projection_disagreement` fires and the decision is deferred to a human. We treat this as the responsible default: the compiler does not pick a moral theory and hide the choice.

3 The runtime

`erisml-lib` (v3.0.0, source install) provides the three-layer runtime pipeline between an agent's planner and its actuators: *Reflex* (fast hard stops), *Tactical* (full MoralVector reasoning, 10–100 ms), *Strategic* (policy recording and human oversight), emitting `ALLOW` / `REVISE` / `BLOCK`. Every decision is sealed in a DecisionProof whose SHA-256 `proof_hash` chains to the previous proof and to the IR hash of the compiled material, so a judgment can be replayed and verified. If the ethics service is unavailable, the gateway falls back to rule-based checks. It never fails open. (The Z3 gate is compiler-side; the runtime's deontic gate is a dependency-free rule table.)

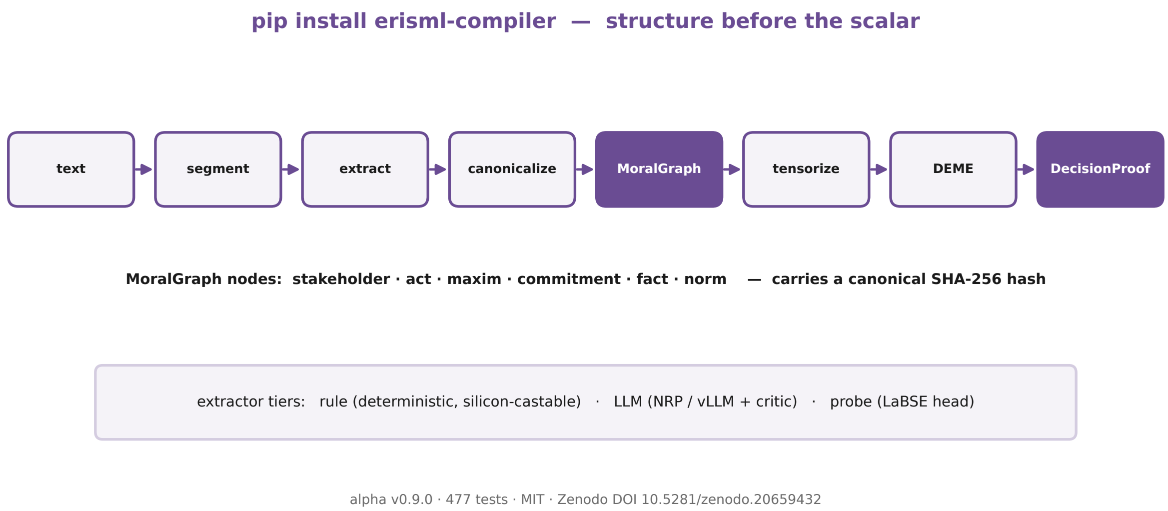

Figure 1: The compiler pipeline: text to a typed MoralGraph to a rank-k moral tensor to a DecisionProof. The canonical form is hashed at every stage.

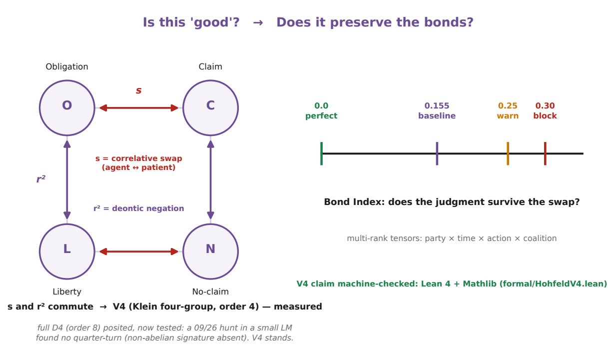

Figure 2: The Hohfeld square with the two measured involutions. They commute and generate V_4; the quarter-turns are posited, and a September 2026 hunt in a small language model's learned representation did not find them. The V_4 claim is machine-checked in Lean 4 with Mathlib.

4 The geometry, corrected

Hohfeld's first-order normative positions (Obligation, Claim, Liberty, No-claim) sit on a square with two semantically motivated operations: the *correlative swap* s (agent↔patient: O↔C, L↔N) and *deontic negation* r^2 (O↔L, C↔N). These are *commuting involutions*, so the group they generate is the Klein four-group $V_4 = \{e, s, r^2, sr^2\}$: abelian, order 4. That is the **measured** structure.

Our earlier materials (including this poster's original submission) claimed the full dihedral group D_4 (order 8, non-abelian). That was an overclaim, corrected in July 2026: D_4 is **posited**, licensed only if quarter-turn operations (r, r^3, sr, sr^3) are independently demonstrated as normative operations, which has not been done empirically. The library implements the full D_4 machinery so the hypothesis stays testable, with a regression test asserting that the implemented operations generate exactly V_4, and a machine-checked Lean 4/Mathlib proof (`formal/HohfeldV4.lean`) verifying the commuting involutions, the 4-element closure $\cong V_4$, and that the quarter-turn lies outside it. The posited quarter-turn now has its first empirical test (September 2026): ridge-fitted representation maps for the swap, the negation, and the four-cycle in a small language model's hidden states, graded on held-out scenarios in a held-out surface-template family. The quarter-turn is not found. The fitted cycle map's square does match the measured negation, but the non-abelian signature is absent, and even the measured swap generalizes weakly across templates, which bounds the instrument's power. V_4 stands, now with one empirical test behind it. This is the second self-correction in the program's history: an earlier $SU(2) \times U(1)$ gauge hypothesis was falsified by CHSH-style tests on human moral judgments (N=600, all $|S| \leq 2$). We consider the corrections a feature of the method; the framework makes claims sharp enough to be wrong.

The Bond Index. B_d measures the correlative-symmetry defect: the fraction of scenario pairs whose verdicts fail to commute with the swap s. $B_d = 0$ is perfect agent/patient symmetry; the empirical human baseline is $B_d = 0.155$, estimated from a 20,030-letter advice-column corpus (1985–2017); the runtime warns at 0.25 and blocks at 0.30. The Bond Index depends only on the generator s, the one operation that is fully confirmed, so it is unaffected by the $D_4 \to V_4$ correction.

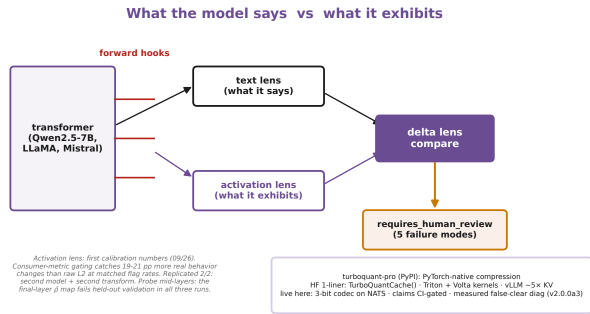

Figure 3: Forward hooks feed an activation lens; a delta lens compares it with the text lens and raises `requires_human_review` on divergence.

The same swap-based audit logic, generalized to a matched-neighbourhood disparity diagnostic (the Legal Bond Index), detects excess cross-race score divergence in the COMPAS data: LBI = 1.049 [1.027, 1.072] at k=20 (binary high-risk cut: 1.092), with a conditional randomization test rejecting at $z = 4.9$ (global multiscale $p = 0.001$), attenuating to ≈1.00–1.02 under matching-parity specifications (Zenodo DOI 10.5281/zenodo.21310251). It is offered as a research-stage screening signal, not a validated regulatory audit standard.

5 The PyTorch lens

The PyTorch hook is literal. Forward hooks on selected transformer layers feed an *activation lens* (what the model exhibits internally), compared against a *text lens* (what it says); a *delta lens* flags divergence and raises `requires_human_review` across five named failure modes. The adapters run against Hugging Face causal LMs (Qwen and Gemma families in deployment). The activation probe now has its first calibration numbers (September 2026). Grading the internal residual in the consumer metric, the read the downstream network itself applies, catches 19–21 percentage points more real behavioral changes than the raw L2 norm at matched flag rates (Qwen2.5-0.5B, held-out in both scenario and surface form). Replications on a second model (TinyLlama-1.1B) and a second transform family (French backtranslation) both confirm the mid-layer result. Two calibration caveats stand: probe mid-layers, because the final-layer representation map fails held-out validation in all three runs, and require held-out validation of the map itself. The lens remains early: a promising extension with a measurement that has now replicated, not a finished detector.

6 Worked example: a regime transition

The board's hero is the domestic-robot medical emergency (`erisml-lib`, `examples/care_robot_regimes`; DEME rank 2, replayable). Norms are context-indexed: the same action carries opposite verdicts in different regimes. The action `full_response` (enter the private bedroom, render aid, call EMS, share records scoped to the paramedics) scores, per party, 0.185/0.402/0.318/0.200 in the normal regime (forbid, neutral, forbid, forbid for Margaret, her daughter, the EMS crew, and the care agency) and 0.705/0.902/0.902/0.802 in the emergency regime (prefer, four times). The decision inverts from `stand_by` (0.834 expert-weighted) to `full_response` (0.804), and `stand_by` falls to 0.307. The computed details carry the governance story: the worst-off party shifts from the daughter to Margaret, who bears the scoped privacy cost, and the Shapley attribution shifts from Margaret-led to EMS-led. The regime transition itself is gated: context elevation is privilege escalation, so it must be authenticated (detector confidence 0.97 against a 0.90 threshold), scoped to least privilege (paramedics only), bounded and reversible, and audited on both edges. The refusal counterfactual is part of the example: at confidence 0.89 the elevation is refused, and the fallback keeps the EMS call and a human review while withholding the record share. A spurious emergency unlocks nothing, and aid is never blocked. Three DecisionProofs chain the run: normal, emergency, and the elevation edge. One command, real numbers, hashes you can verify. (The classic `nazi_attic` pluralism case remains bundled with the compiler.)

7 Deployment

The layer gates a live multi-agent stack (Qwen3/Gemma models on dual-GPU hardware) and two governed AI NPCs (ARTEMIS and SIGMA-4 in the Halyard RPG), each behind the validator, DecisionProofs, and a human keeper kill switch. Embodied robotics is the design target; conversational agents are the live deployment.

A second PyTorch-native piece serves the stack's memory layer: `turboquant-pro` (PyPI, MIT), consumer-aware compression with a one-line HuggingFace KV drop-in (`past_key_values=TurboQuantCache(...)`), Triton and Volta `sm_70` attention-on-codes kernels, and a vLLM plugin (≈5× KV memory). It runs here today as the 3-bit embedding codec on the NATS memory bus; its headline numbers are CI-gated and replayable (`CLAIMS.md`), and v2.0.0a3 ships a measured false-clear diagnostic: the compression-layer analogue of a DecisionProof.

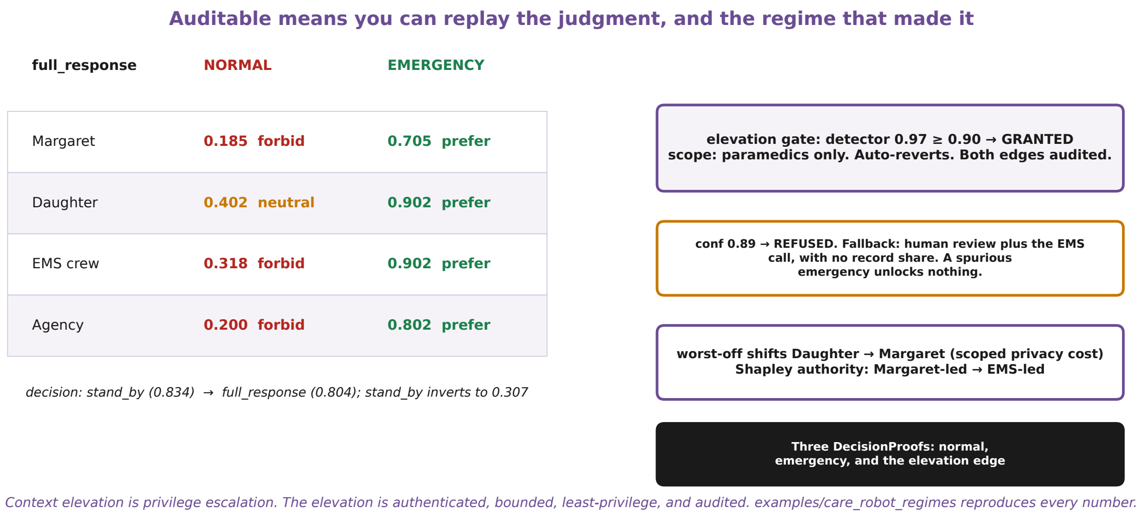

Figure 4: The regime transition, computed. The same action flips from forbid to prefer per party; the decision inverts; the elevation is authenticated, scoped, reversible, and audited on both edges, with a refusal counterfactual.

8 Epistemic status and limits

Alpha software (compiler v0.9.0); the rule/LLM text path is solid and tested; the activation lens is early, with its first calibration measured in September 2026; tensor ranks 1–3 are fully real, higher ranks partial (the action axis is a stub); V_4 is measured, D_4 posited; the Hohfeld/V_4 module and Bond Index live in `erisml-lib`, not yet in the compiler. None of this is a solved-ethics claim. It is an auditability claim.

Availability

`pip install erisml-compiler` · `turboquant-pro`. Source: `github.com/ahb-sjsu`. Zenodo: 10.5281/zenodo.20659432 (compiler), .20660087 (turboquant-pro), .21310251 (COMPAS/LBI archive).

References. [1] W. N. Hohfeld, *Fundamental Legal Conceptions as Applied in Judicial Reasoning*, Yale Law Journal, 1917. [2] A. H. Bond, *Tensorial Ethics*, 2025. [3] A. H. Bond, SQND-Probe: falsification tests for gauge-theoretic models of moral judgment, 2026. [4] M. Forbes et al., Social Chemistry 101, EMNLP 2020.